2026 Active Inference Symposium — Presenter Program

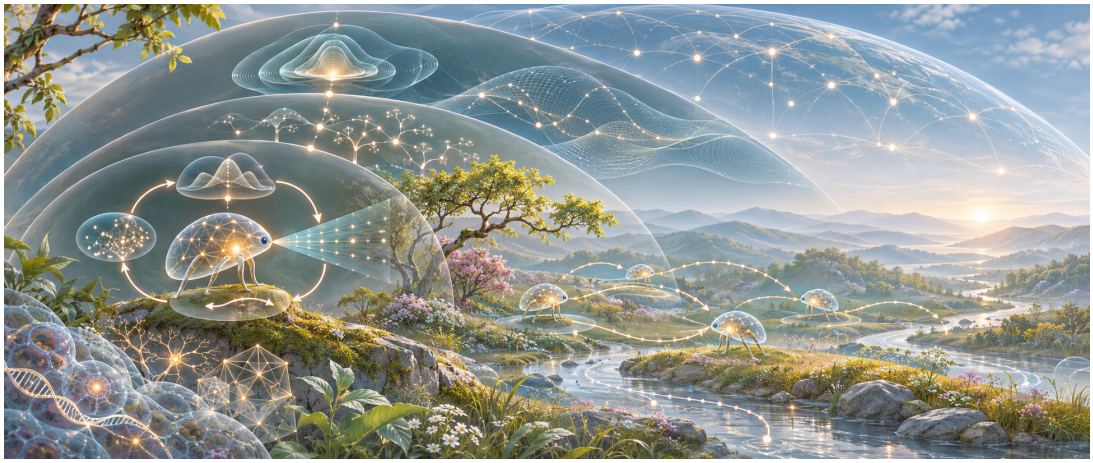
Working Program

November 12–13, 2026 (UTC)

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Active Inference Institute

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Book of Abstracts



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About this program

Event dates: 2026-11-12 to 2026-11-13.

This working program collection contains approved public abstracts and explicitly marks entries awaiting an abstract. Inclusion does not assert that a presentation has occurred or that a recording has been received.

Presenter details reflect public self-reports; unavailable fields are omitted.

1 An in-depth discussion on “World models in natural and artificial intelligence”

Presenter: Bradly Alicea

Contribution: Interactive Workshop

Affiliations: Active Inference Institute (SAB)

Abstract

This session will be a summary and discussion group for the special issue ”World Models in Natural and Artificial Intelligence”, published in the Royal Society A journal in May 2026. We will offer a summary of the special issue content, along with a discussion of Active Inference related topics and future directions from the work presented in the issue.

2 Build a curious rocket: a hands-on continuous-state AIF session with cpomdp

Presenter: Dan Corva, Kevin Backhouse

Contribution: Interactive Workshop

Abstract

In the 1960s, Rudolf Kalman wrote down a filter that would go on to navigate Apollo to the moon. It has been the workhorse of continuous-state estimation ever since, and if you have ever built a continuous active inference agent in the linear-Gaussian regime, you have built one too, whether you intended to or not. Bolt a goal onto it and you get a very good controller that is not curious in the slightest. Koudahl, Kouw and de Vries proved why in 2021: under a fixed sensor the epistemic value is identical for every action, so expected free energy quietly reduces to Linear Quadratic Regulator (LQR). In this workshop, in about ten lines of Python that will look familiar to anyone who has used pymdp, we are going to build a rocket! Then, we add the interesting part, curiosity, by making a surgical, biologically realistic change: adding sensor noise that varies based on the rocket's position. We will see how the flight path bends when we add artifacts like a navigation beacon or a blackout zone, and explore the science as to why. You will write every line yourself in a Colab notebook and do not worry there will be plenty of scaffolding from the cpomdp team. Code hints will drop in on a timer, and a reference cell for anyone who gets stuck or would rather just see the finished steps. We will build this in three steps: the 1960 rocket, a nav beacon it can fix its position by, and a blackout zone. Everything we use runs on the PyPI release of cpomdp and anything you build in the notebook is yours to keep and continue working on. Prerequisites: comfortable editing a few lines of Python. Familiarity with pymdp helps but is not required. Nothing to install; a browser and a google account is enough. If you would like to use your own IDE feel free but we may not be able to provide 1-on-1 support if you get stuck.

3 A Modular Perspective Layer for Active Inference Agents

Presenter: Hongju Pae

Contribution: Interactive Workshop

Abstract

Most agent architectures collapse two things that phenomenology keeps apart: what the agent should do immediately, and what kind of world the agent takes itself to be in. This session introduces “perspective” as a distinct architectural layer that carries the second. Perspective is a slowly evolving, history-sensitive latent state that gates how observations are encoded. Importantly, it is decoupled from policy gradients and is never itself an optimization target. Because it is decoupled, it is portable and can be attached on top of an existing policy-optimizing agent rather than requiring a bespoke design. The first part of the session develops the conceptual frame: how a phenomenologically motivated condition translates into a concrete architectural commitment, and why that commitment is orthogonal to belief-state inference and policy selection. I then trace the capabilities that this single layer has been shown to support across a sequence of studies, including directional hysteresis, history-dependent perceptual reorganization, and body-grounded conative orientation, treating these as unified aspects of a single portable module. There will then be a live demonstration attaching the perspective layer to an active inference agent, such as a hierarchical Bayesian / discrete state-space model, and running it in parallel with its unaugmented counterpart in the same environment, so participants can see exactly what the layer adds and where it plugs in. The session closes with a structured discussion in which participants are invited to identify candidate insertion points for a perspective layer in their own active inference agents, whether built in PyMDP, RxInfer, or custom frameworks. The aim is to leave participants able to reason about whether and how a subjective perspective could be added.

4 Active Inference and Free Energy Principle in 2026

Presenter: Karl Friston

Contribution: Keynote & Panel

Abstract

No program abstract has been published for this presentation yet.

5 Active Inference, Bayesian Causal Learning and Empowerment in RL.

Presenter: Alison Gopnik

Contribution: Livestreamed panel or group session

Abstract

Cognitive scientists have applied advances in our formal understanding of causation in philosophy and computer science, particularly within the Causal Bayes Net formalism, to understand human causal learning. This formalism is also closely related to ideas in active inference. In parallel, reinforcement learning researchers have developed the idea of an intrinsic reward signal called “empowerment”. An agent is rewarded for maximizing the mutual information between its actions and their outcomes, regardless of the external reward value of those outcomes. “Empowerment” may be an important bridge between classical Bayesian causal learning and reinforcement learning and may help to characterize causal learning in humans and enable it in machines

6 Root Frequency Theory: An Integrative Framework for the Continuity of Lived Experience

Presenter: Bianca Avanzo

Contribution: Livestreamed panel or group session

Abstract

How do we keep a coherent sense of self across time — and can that process be measured in real time, on consumer hardware, before it breaks down? Root Frequency Theory (RFT) proposes that experiential continuity is not a default state but an ongoing biological achievement requiring coordination across five nested regulatory layers: environmental constraints (C0), autopoietic regulation (C1), predictive neural integration (C2), narrative self-organization (C3), and the continuity of lived experience (C4). Fragmentation is defined as a self-reinforcing cascade of cross-scale misalignment. The cascade is bidirectional: structured self-knowledge introduced at C3 — operationalized as epistemic alignment (K_e) — is hypothesized to initiate upward regulatory stabilization. To test this, I developed the M-RFT metric: $M = (\Phi \cdot K_e) / F$, where Φ indexes autonomic flexibility via HRV, K_e composites interoceptive accuracy and narrative coherence, and F captures regulatory burden. I then built Root Extension — an iOS application using iPhone and Apple Watch — to administer this protocol ecologically and deliver personalized epistemic scaffolding upon session completion. The N=1 longitudinal study ran 321 sessions across 23 active days (April 16–May 14, 2026), with the researcher as the sole subject. Key findings: mean M = 3.79 (Mdn = 2.14), intra-session regulatory recovery — FRAGMENTED to RESILIENT within the same calendar day — observed on 34.8% of active days. This presentation introduces the theoretical architecture, the metric, the app protocol, and the preliminary data — and invites the active inference community into the next phase: a formal N=20–40 validation trial planned for 2027.

7 Working Groups Sessions (2 sessions: Opening & Charter; Outcomes & Next Steps)

Presenter: Ed Ober

Contribution: Livestreamed panel or group session

Abstract

AII proposes to facilitate / host active inference working groups as explained in our paper “Coordinating the Field”, with a field group and one or more domain groups. The first symposium session (Opening & Charter, November 12) will introduce the concept, recap the paper and reason for the groups, and solicit suggested domains and participants. It will also review the proposed charter and consider amendments, with versioning per group, considering nominations for chair(s) and secretary(ies), with a goal of adopting the charter(s) during the symposium if possible and establishing post-symposium meeting cadence and work plan. The second session (Outcomes & Next Steps, November 13) will report back to the main session, recap what was accomplished during the symposium and announce any working groups formed and their first year goals established.

8 CogNarr Story Graph: A Formal Semantics for Narrative as a Belief Model

Presenter: John Boik, Glen EP Ropella

Contribution: Livestreamed panel or group session

Abstract

People naturally express their beliefs about a topic, situation, or problem in the form of narrative. CogNarr is a soon-to-be-released, open-source, interactive tool that lets a writer transform narrative into an explicit, formal semantic graph, which they can inspect and correct until it faithfully reflects their belief model. A CogNarr story graph captures a narrative's argument structure, its timing, and the modality of its claims, including predictions and conditionals, together with the uncertainty attached to each. In general it expresses the verifiable meaning of what a person wishes to say, while reducing the ambiguity of natural language. As such, it could be of use in modeling the beliefs that active-inference processes operate over, and as a framework within which active-inference methods could be employed to infer semantic meaning from text. Because it makes a person's beliefs explicit, transparent, and shareable, CogNarr has potential wherever consequential beliefs are expressed as narrative, from medicine and governance to public polling and accident reporting. As well, CogNarr story graphs support accurate agent-agent communication. The collective-cognition aim behind the project is alignment of shared belief models during decision-making processes in the large-group setting. The talk will be a preview of features and capabilities, with some discussion of potential applications.

9 Active Inference and the Evolution of Mind and Society

Presenter: Luca Possati

Contribution: Livestreamed panel or group session

Abstract

This panel explores how active inference and the free-energy principle can contribute to a unified account of psychological and social evolution. It examines how cognition, affect, social interaction, cultural practices, and institutions emerge through processes of prediction, adaptation, learning, and uncertainty reduction. By bringing together perspectives from cognitive science, philosophy, anthropology, and social theory, the panel asks how individual and collective forms of agency co-evolve and how shared models of the world shape both minds and societies.

10 From Self-Evidencing to Global Governance: Active Inference and the Dynamics of Justice

Presenter: Majid D. Beni

Contribution: Livestreamed panel or group session

Abstract

Active inference frames biological and cognitive agency as the imperatives of self-evidencing—maintaining systemic integrity by minimising variational free energy within an open, stochastic environment. Under this formalism, agents, from single cells to complex organisms, persist by regulating their internal states against entropic dissipation, enacting a Markov blanket that partitions self from world. This imperative entails sophisticated inference, balancing epistemic drives against pragmatic preferences, thereby constituting adaptive agency. This talk extends this foundational picture of selfhood to the domain of political communities and global justice theory, demonstrating that the same dynamical constraints that govern individual persistence scale hierarchically to structure collective organisation and inter-state conduct. Drawing on the Free Energy Principle, political communities are modelled as open, self-organising systems embedded within a nested hierarchy of interacting Markov blankets. Their stability is not intrinsic but relational, contingent upon continuous exchange of information, resources, and institutional constraints with their environment. Within this multiscale architecture, global justice emerges as an active regulatory imperative: a homeostatic constraint that maintains a nonequilibrium steady state across coupled societies. We formalise this through the expected free energy decomposition, where just policies balance risk—adherence to shared normative constraints such as human rights and non-aggression—against ambiguity—the active resolution of uncertainty arising from structural interdependence. The normative implications are acute. We identify a pervasive pathology of self-organisation, wherein dominant or insular political regimes mis-specify themselves as closed systems. By suppressing external coupling and attenuating sensory evidence, they optimise local coherence at the expense of systemic stability. Drawing parallels to oncogenic cellular proliferation and psychotic cognitive inference, we argue that such closure systematically underestimates global risk and propagates destabilising volatility. Ultimately, we contend that justice is not a static distributive outcome nor a single-level contractual procedure, but the dynamic property of a multiscale system that successfully synchronises its trajectories under persistent uncertainty. This offers a unified, first-principles framework for linking the philosophy of agency to the normative demands of a deeply interconnected global order.

11 “Evolution, Emergence, Prediction.”

Presenter: Mike Levin

Contribution: Livestreamed panel or group session

Abstract

No program abstract has been published for this presentation yet.

12 A New Politics: From Linear to Sorted Politics Through the Policy Map

Presenter: Montijn van Loo

Contribution: Livestreamed panel or group session

Affiliations: Tilburg University

ORCID: 0009-0003-0249-3240

Presenter background

I am an econometrist gone rogue. Instead of throwing mathematics at finance, I throw Active Inference mathematics at philosophy. See for the consciousness, morality, politics, gender, free will and God applications, with the poem uniting them all.

Abstract

Political disagreement often reflects competing worldviews: different beliefs about how the world works and what will happen under alternative policies. Yet political debate proceeds linearly, through proposals, responses, and counterresponses. The underlying reasoning and evidence remain fragmented, assumptions can escape scrutiny, and criticism loses its target as debate moves on. Consequently, incompatible worldviews can persist without their disagreements being resolved. This talk proposes a new politics, motivated by Active Inference, which provides the mathematical object around which political debate can be organized: the policy map. This map expresses a worldview through predicted outcomes under alternative policies. Making it the persistent object of debate allows criticism to retain a stable address rather than disappear in the rhetoric of successive exchanges. The mathematics of the policy map distinguishes three types of criticism: belief criticism, challenging beliefs about current states or the consequences of policies; outcome incompleteness, identifying omitted consequences; and policy incompleteness, identifying missing alternatives. These provide the horizontal sorting of political debate. Vertical sorting organizes the reasoning beneath each prediction, allowing criticism to reach its underlying assumptions and evidence. Unresolved objections remain visible until answered or incorporated through revision, potentially changing the predicted outcomes and preferred policies. In this way, Active Inference offers a normative application that goes beyond describing politics as if political actors performed Bayesian inference. It motivates transforming the very structure of political debate, replacing rhetorical exchanges with cumulative criticism and incentivizing a shared search for the worldview that best fits the available evidence. The ultimate aim is a politics in which criticism can actually bite, transforming not only political debates, but the world shaped by its decisions. For the only path forward is critique without exception.

13 Explicating and Comparing Prediction Architectures of Theories of Consciousness and its Application to INTREPID

Presenter: Montijn van Loo

Contribution: Livestreamed panel or group session

Affiliations: Tilburg University

ORCID: 0009-0003-0249-3240

Presenter background

I am an econometrist gone rogue. Instead of throwing mathematics at finance, I throw Active Inference mathematics at philosophy. See for the consciousness, morality, politics, gender, free will and God applications, with the poem uniting them all.

Abstract

Consciousness: The Channel Framework and its Application to INTREPID The Channel Framework treats theories of consciousness as relating physical and experiential events, with the minimal relation being predictive. The Channel Framework embeds this psychophysical relation within the full experimental prediction architecture, connecting physical processes, conscious content, introspective access, and observable outcomes. Expressed through probabilistic channels, these architectures can be interpreted as generative models of experimental predictions, providing a common language in which different theories can be explicated and compared. I apply the framework to INTREPID's adversarial comparison of Integrated Information Theory and predictive-processing theories of consciousness, including Neurorepresentationalism and the Active Inference theory of consciousness (AI-C). The framework asks whether their prediction architectures can be faithfully aligned, which theoretical differences actually generate their contrasting experimental predictions, and whether these differences carry the particular claims the experiments aim to test. Building on this analysis, the Channel Standard develops criteria for attributing evidence from competing theory predictions to specific theoretical claims. The broader aim is to move beyond theory-level comparison toward methodological concensus.

14 Recent advances in empirical applications of active inference in computational psychiatry

Presenter: Ryan Smith

Contribution: Livestreamed panel or group session

Abstract

In this talk, I will review recent studies and results using active inference and related modeling approaches to study mechanisms of normative cognitive-affective processes and associated neural responses, and how these appear to be affected in different psychiatric conditions. I will also discuss contexts in which active inference appears to offer potentially unique resources for modeling these processes. Conversely, I will discuss contexts in which active inference may overlap with simpler modeling approaches and related issues centered around the level of mechanistic description one is interested in studying.

15 An elementary form of agency at planetary-scale

Presenter: Sergio Rubin

Contribution: Livestreamed panel or group session

Abstract

The claim that climate homeorhesis—non-equilibrium steady-state dynamics that underlie Earth's long-term habitability—depends on an integrated biosphere–climate system capable of acting on its own is central to debates surrounding the Gaia hypothesis. This paper presents a proof of concept suggesting a minimal form of planetary agency based on a simple biosphere–climate system model operating through active inference. This means that the internal states of the system (biosphere) will appear to infer shifts in net incoming solar radiation (external states) through its boundary states, i.e. by integrating past and current surface temperatures (sensory states) and actively changing greenhouse forcing (active states) to maintain climate trajectories within geophysiological bounds. When surface temperature is altered by solar fluctuation changes, the system adjusts greenhouse and albedo parameters to restore expected climate trajectories, analogous to correcting sensory prediction errors in biological systems. Perturbing active states (e.g., imposed changes in albedo) does not reduce inference precision. Inference robustness decreases when the system's degrees of freedom increase. Active inference falls, and thus climate homeorhesis fails when there is no biosphere activity (lifeless planet) or when it is decoupled from the climate system. Predictive behaviour operates within the range of solar constants consistent with life's presence on Earth (~ 4 Ga to ~ 1.5 Ga expected in the future); outside this range, inference fails, undermining a putative form of planetary agency.

16 Reverse engineering generative models from zebrafish decision-making

Presenter: Takuya Isomura

Contribution: Livestreamed panel or group session

Abstract

Understanding biological intelligence requires identifying the variational invariant structures underlying neuronal dynamics. Experimental data provide access to the evolution of neural activity and synaptic weights. These dynamics can be expressed in different coordinates while preserving a common dynamical organisation. In Bayesian coordinates, they can be read as belief updating under an implicit generative model; in computational coordinates, as action generation, including behaviour, decision-making, and algorithmic computation. Triple equivalence formalises transformations among these descriptions and identifies their shared variational structure. The empirical question is which implicit generative model is instantiated by a given neuronal system. Here, reverse engineering was applied to neural and behavioural data from zebrafish learning a visually guided go/no-go decision-making task. From initial neural activity, the method identified individual-specific implicit generative models and instantiated them as synthetic agents that minimise variational free energy. These agents generated trial-by-trial predictions of subsequent synaptic plasticity, neural activity, and behavioural performance. The predictions captured individual learning trajectories and generalised to previously unseen task conditions. Their agreement with unseen empirical data provides a quantitative test of the predictive validity of the free-energy principle in behaving animals. More broadly, this framework shows how present neuronal states can be used to forecast future learning and self-organisation in individual biological systems.

17 Knowthyself — A code-free active inference workbench

Presenter: Ted J. Vrioni

Contribution: Livestreamed panel or group session

Affiliations: Junior Consultant at Vrioni Consulting

Presenter background

physics, cognition, computation, art — my vibe tends to be more of a generalist, although music is really what I really peak at in terms of “performance”. I see it as “engineering consciousness”, but in the first person pov… and it’s not about creating signal: it’s about filtering noise. Dances of symbols all along. So the work is internal so far, but it is trying to find a way out. I am also working on a few papers though, about physical agents (involving Chladni patterns), making physics with stamps (involving dynamic causal modelling and “ignorance modelling”), consciousness (involving implications of Noether’s theorem for electromagnetic waves), and philosophy (translating Bertrand Russell’s Nobel Lecture’s lessons into active inference exercises — his speech is on Youtube, you’ll love it). It’s private and solo-work so far… slowly but surely.

Abstract

If pymdp was an engine, Knowthyself is an attempt to give it a chassis and wheels. The whole point is to promote a workbench to build your active inference agent and environments without writing a single line of code. It’s beginner- and expert-friendly, fostering education, research and development, and maybe the idea of a “hub of digital twins” for currently existing literature. It comes in progressive levels of “difficulty” which unlock various tools and functionalities. You become a particular chemist, authoring all sort of “herds of strange things”.

18 A Lexical T-Maze for Active Inference: A Proof-of-Concept Model of Context Seeking Under Ambiguity

Presenter: Ahmad Nazzal

Contribution: Pre-recorded Video presentation

Affiliations: Zayed University

ORCID: https://orcid.org/0000-0002-6558-0734

Abstract

Lexical-disambiguation tasks with supplied context do not reveal whether readers would choose to seek evidence before deciding. Therefore, we adapted the spatial T-maze in the active-inference package pymdp to show how agents seek and use contextual information under lexical uncertainty. Agents could choose between two readings or request a contextual hint. We performed two main sets of simulations. First, deterministic simulations used fully informative hints to reproduce context seeking followed by lexical commitment. A hint shifted belief over two readings from equal probabilities (0.50, 0.50) to certainty (1.00, 0.00), reducing entropy by one bit. Second, lexical decision trials were repeatedwith informativeanduninformative hints across 32 paired simulationruns. Agents learned the relationship between the intended reading and the observed hint. Informative experience increased both the probability and simulated frequency of requesting context. This increase persisted when trials ended after one response and feedback. A hint information-gain ablation, in which policies were rescored without crediting the information expected from hints, eliminated the learning-related increase in hint-request probability. These simulations demonstrate how experience changes the expected informational value of context within a lexical task, providing a computational proof of concept for subsequent behavioral experiments.

19 FEP and Active Inference through the lens of Winnicott's Potential Space

Presenter: Alexander Sabine

Contribution: Pre-recorded Video presentation

Affiliations: Active Inference BoD

ORCID: https://orcid.org/0000-0001-8462-2656

Presenter background

CRR. Change has its own clock. Empty Now. Equanimity. Continuous Learning. AI safety

Abstract

Winnicott held that the infant's transitional object must never be questioned - did you create it or find it? The paradox is to be held, not resolved, because play, creativity and culture live in the space it opens. This talk argues that active inference carries the same paradox. Perception is jointly authored by the generative model and the evidence, and action shapes the world the agent then finds. I draw thinking from the pioneers of early childhood education, philosophy of time and contemporary discussions. I explore some possible consequences, treating the unresolved paradox as a design principle.

20 Active Inference Tutorial: Perspectives, Correlates, and Future Directions for Graphical Brains

Presenter: Andrew Pashea

Contribution: Pre-recorded Video presentation

Affiliations: Active Inference Institute (SAB)

Abstract

This Symposium 2026 tutorial covers Active Inference from various angles based on the experience over several years with teaching and applying Active Inference. It begins with a brief descriptive rehearsal of fundamental ideas and principles akin to other previous tutorials to establish a foundation for the audience. However, this will be followed by a tour through various textbook materials and publications, past and present, to emphasize crucial areas for understanding modelling and application areas. This tutorial will include visual and discursive illustrations of a range of model classes for different use-cases; discussions of existing and under-development software libraries; and some frontiers in modelling for identifying directions for further research applications which can be especially valuable for the broader research community in 2026 onward. Topics will include, non-exhaustively, the Bayesian brain hypothesis and behavioralism versus partial observability and modelling internal states; principles of exploration, learning, and novelty; fundamental discrete and continuous models in Active Inference and their relationship; computational psychiatry and its significance to other domains; the FEP and its historical and practical relationship to Bayesian model comparison; software libraries including SPM, pymdp, RxInfer, RGMs, and other developments; hierarchical and hybrid models; and structure learning.

21 Inferring the Unseen

Presenter: Anna Pereira

Contribution: Pre-recorded Video presentation

Abstract

As AI systems increasingly mediate access to scientific knowledge, how they communicate complex theories becomes consequential for public understanding. Active inference presents a distinctive challenge: its mathematical and computational foundations offer explanatory resources, yet accessible accounts must distinguish formal results, empirical evidence, and emerging interpretations. Simplification can support understanding, but can also obscure uncertainty or encourage misleading conclusions about personal experience. Building on Pereira and Chamberlin's manuscript, this foundational study will explore recommendations for large AI companies communicating active inference to general audiences. Its central question is how AI communication can preserve scientific integrity while helping people understand both the framework's core concepts and their potential relevance to everyday life. Particular attention will be given to the difference between being able to repeat an explanation and developing an understanding that supports reflection, judgment, and appropriate application. The project will draw on literature spanning active inference, science communication, human–AI interaction, and learning. It will examine candidate communication approaches, including layered explanations, everyday examples, explicit statements of uncertainty, and prompts that connect abstract concepts to users' existing knowledge. A potential evaluation framework could consider conceptual accuracy, accessibility, perceived personal relevance, and the ability to distinguish supported implications from speculation. This framework could guide empirical evaluation of approaches. The proposed contribution is an initial set of communication principles and research questions for AI developers, researchers, and educators. Rather than assuming that greater accessibility necessarily produces better understanding, the study will examine the tensions between clarity, scientific fidelity, personal relevance, and unwarranted certainty. The broader aim is to support communication that enables people to engage meaningfully with complex science without presenting a theoretical framework as a comprehensive explanation of their lives. This thirty minute presentation will outline the rationale, proposed framework, and recommendations emerging from the work, alongside priorities for further research and evaluation.

22 A precis of two principles - Towards a relational perspective on the FEP

Presenter: Avel Guénin—Carlut

Contribution: Pre-recorded Video presentation

Abstract

The Free Energy Principle offers a physically grounded account of intentional properties in mind and life, framed within Dynamical Systems Theory. Because of this generality, it was proposed as a formal basis to study biological autonomy, autopoiesis, and enactive cognition. Yet, those approaches are defined by a relational understanding of mind and life, centred on the description of engagement with the world as a constructive process where agents are "laying down a path in walking". Building from Rosen's relational biology, we argue that such processes are by construction outside a "Newtonian" description in terms of well-defined observables and their dynamics. Rather, we draw from Fields' demonstration of the emergence of contextuality – the participatory production of observables against which the agent regulates their activity – in classical, thermodynamically bounded processes of observation. We suggest a complementarity between Friston's classical FEP and Fields et al.'s "quantum" (or, in our account, relational) FEP, describing respectively the dynamics within a biological / cognitive space of relevant observables and the physical processes underlying the construction of that space. Finally, we propose a structural realist interpretation of this complementarity within the "closure of constraints" account of biological autonomy, where both principles are understood to act on a physically embedded system of biological constraints rather than an abstract mathematical space assumed to fully represent biological / cognitive organization. This approach foregrounds the synthesis between mathematical and theoretical approaches to biological autonomy, thus supporting a principled approach to both analyzing the organization of natural systems and attempting the implementation of artificial autonomy.

This presentation is adapted from Guénin--Carlut, Avel. 2026. "A Tale of Two Principles - Towards a Relational Perspective on the FEP." https://hal.science/hal-05697510 ? ., currently under review at the Entropy Special Issue "Active Inference in Cognitive Neuroscience".

23 Maladaptive Daydreaming as a Precision Trap: An Active Inference Framework, Its Testable Core, and Evidence from Daily Diaries

Presenter: Bo Lee

Contribution: Pre-recorded Video presentation

Abstract

Maladaptive daydreaming (MD) involves prolonged immersion in self-generated fantasy and difficulty disengaging despite awareness that the experience is imagined. We propose a testable active-inference framework in which the precision assigned to internal predictions relative to external sensory evidence may contribute to self-sustaining fantasy. This precision-feedback mechanism is a hypothesis, not an established clinical explanation. Using analytic counterexamples and 115,200 synthetic datasets, we show why finite-time persistence and hysteresis-like observations alone do not identify bistability, and examine how ramp-and-hold experimental designs can improve discrimination among competing models. Exploratory secondary analyses of daily-diary data are discussed at their actual timescale, without treating daily associations as proof of an episode-level mechanism. The talk connects active inference, computational psychiatry, experimental design, and open science. It concludes with falsifiable predictions and a research agenda focused on measuring internal and sensory precision, tracking within-episode dynamics, comparing alternative models, and testing feasible interventions. We welcome discussion of practical measurements and alternative explanations.

24 The Benefits of Circuit Convolution: behavioral complexity arises from developmentally- simulated nonlinearity.

Presenter: Bradly Alicea

Contribution: Pre-recorded Video presentation

Affiliations: Active Inference Institute (SAB)

Abstract

While the generation of behaviors is generally substrate-agnostic (there is more than one way to build a neural circuit), complex behaviors require an increase in representational complexity. We explore an architecture called the biological Rube Goldberg Machine (bRGM). This will be used to model the Developmental Neurosimulation of a generalized architecture that produces spatial navigation behaviors. Developmental Neurosimulation has been used to simulate the development of Braitenberg Vehicles. In this talk, we will explore the neurosimulation of Cybernetic Convolution Networks (CCNs), which have the potential to add great complexity to sensor-effector pairings. Additional connections to agentive learning and optimization models of behavior will also be discussed.

25 The Minimal Necessary Basis for Primitive State-of-Affairs Inquiry

Presenter: Bruce Tisler

Contribution: Pre-recorded Video presentation

Abstract

Discovering whether there is a minimum structure required for inquiry to be complete. The paper begins with six familiar questions: Who, What, Where, When, How, and Why. But the research is asking something deeper: Are these six categories simply a useful convention, or does each represent something fundamentally necessary about what can be known? Consider the bridge-collapse example. Two bridge collapses occur. For both, the involved parties are identified (WHO: the same engineering firm and inspection authority in both cases). The event is specified (WHAT: structural failure of a suspension bridge). The locations are of the same type (WHERE: bridges of identical design crossing rivers of similar flow rate). The time and sequence of failure are identical (WHEN: both collapse within a three-minute window following the same triggering load). The mechanical failure sequence is identical in both cases (HOW: the same crack propagation pattern, the same load transfer failure, the same sequence of component failures). Even with all five answers fixed and identical across both collapses, the explanation differs: in the first case, deferred maintenance created metal fatigue that made the bridge susceptible to normal load; in the second, a design flaw in the original specifications made the bridge inherently susceptible under identical conditions. The identical HOW does not determine WHY. The joint answer set does not determine WHY. Therefore WHY is independent of WHO, WHAT, WHERE, WHEN, and HOW.

26 Artificial Sentience Project

Presenter: Cleber Gomes

Contribution: Pre-recorded Video presentation

ORCID: https://orcid.org/0009-0005-9134-2384

Abstract

This project introduces an innovative computational architecture that bridges cognitive neuroscience and interactive simulation by integrating na Active Inference cognitive core directly into UI engines. Grounded in the Free Energy Principle, the system continuously minimizes Variational Free Energy (VFE) to drive autonomous agent behavior, dynamic navigation, and adaptive decision-making within complex, changing environments. Unlike traditional rule-based non-player characters (NPCs) or standard robotic controllers, the agent maintains an internal affective engine that maps multi-dimensional emotional telemetry—such as curiosity, motivation, stress, and fear—directly to behavioral drives, energy management, and rich audio-visual expressions. These affective states are rendered in real time through belief-versus-reality spatial overlays, floating emotional telemetry indicators, and adaptive state vocalizations that mirrors the agent's psychological trajectory. By unifying autonomous perception-action loops with transparent emotional expression, this framework offers powerful capabilities for powering both advanced games and autonomous robots. Its potential applications span multiple domains: Entertainment: Powering next-generation autonomous game characters and dynamic simulations with emergent, lifelike behaviors. Education: Serving as an interactive pedagogical platform for exploring cognitive architectures, cybernetics, and affective computing. Companionship & Emotional Support: Creating empathetic digital companions and social robots capable of contextual awareness and authentic emotional resonance to foster meaningful human-agent interaction and support human well-being. Ultimately, this project demonstrates that Active Inference provides a robust, biologically inspired foundation for artificial systems that not only navigate physical spaces intelligently but also communicate and connect through shared affective experiences.

27 Realms of cognitive engagement in Active Inference

Presenter: Colin Turfus

Contribution: Pre-recorded Video presentation

Affiliations: https://substack.com/@colinturfus

Presenter background

I write a substack blog at where I explore the implications of Active Inference for moral and philosophical issues

Abstract

We consider how Active Inference ideas shed new light on issues in epistemology and axiology, in particular proposing a unifying paradigm based on understanding human enquiry as a process of engagement operating in multiple realms, specifically embeddedness, embodiedness, reflection, participation and immersion. We look at what can be learned from within this framework about areas which have become flashpoints in society such as conflicting worldviews and identity politics.

28 NUMEN: Bare metal inference

Presenter: Daniel Dragolich

Contribution: Pre-recorded Video presentation

Abstract

Come see the beauty of mathematics. Its just not some fancy add on to life. It is life.

29 GNN in 2026: Generalized Notation Notation as a Specification Layer for Applied Active Inference

Presenter: Daniel Friedman

Contribution: Pre-recorded Video presentation

Affiliations: Active Inference Institute (Officer), College of the Redwoods

Abstract

Generalized Notation Notation (GNN) was introduced in 2023 (Smekal & Friedman) as a text-based standard for writing Active Inference generative models, with a “Triple Play” of linguistic, visual and executable renderings. This talk reports where GNN stands in 2026 and what it enables for applied Active Inference. A GNN file is plain Markdown: a state-space block declaring variables and dimensions, a connections graph, initial parameterization (e.g. A, B, C, D, E and related terms), equations, time settings and Active Inference Ontology annotations. The open-source GNN pipeline (v5, 25 steps) parses, type-checks and validates these files, visualizes and exports them, maps them to the ontology, renders them to backends including PyMDP, RxInfer.jl, ActiveInference.jl and JAX, executes the results, and reports each run against source hashes and configuration. The repository’s 38 exemplar models span static and dynamic perception, discrete POMDPs, the epistemic T-maze, continuous and hybrid models, hierarchical and deep temporal models, Dirichlet learning, precision and curiosity, and multi-agent stigmergic swarms. The core is GNN as the specification layer for real projects. The Active Inference Language Model (AILM, released October 2026) uses GNN to specify each layer of a six-level hierarchical discrete language system (letters, phonemes, words, case roles, sentence types, document position), executed in pymdp with no transformer, neural network or learned embedding, and checked with a gate-audit method that tests claims against trivial baselines and deliberately broken variants and ties each reported number to its recorded run. Shorter case studies: deterministic codebase-to-GNN translation (COGANT), inference over assembly traces (Active Assembly), Battleship, and gratitude and model revisability (Thanks for Nothing). Throughout, I distinguish what a GNN declaration establishes (an inspectable, reproducible model) from what needs separate evidence (backend support, numerical acceptance, empirical claims). I close with open problems and invitations for contributors: new exemplars, backends, ontology mappings and teaching materials.

30 Stimming as a Temporary Homeostatic Buffer and Modulator in Autism Spectrum Disorders

Presenter: Diaconescu Adrian-Stelian

Contribution: Pre-recorded Video presentation

Abstract

Repetitive and self-stimulatory behaviors have been re-evaluated over time, evolving from simple dysfunctions to adaptive self-regulatory strategies, however, a unified biological framework capable of explaining their contextual activation and underlying physiology remains incomplete. The present paper proposes a theoretical-hypothetical model (grounded in the Free Energy Principle and the Active Inference framework) that conceptualizes stimming as serving a dual role: both as a temporary homeostatic buffer, activated during states of prefrontal executive overload, and sensorimotor depletion and as a modulator of attentional processes and information processing. Distinctions are delineated between these forms of stimming and the non-creative flow (FNC/NCF) state, alongside specific considerations regarding neurodevelopmental conditions.

31 Partner-Specific Affective Precision in Social Active Inference

Presenter: Harshil Shah, Andrew Pashea

Contribution: Pre-recorded Video presentation

Abstract

In multi-agent social settings, model reliability varies across relationships. Beyond inferring what others will do, an agent must calibrate how confidently those inferences should guide policy selection for each relationship. An agent may maintain a well-validated model of one partner, a fragile model of another, and a model under revision for a third; collapsing these into a single confidence estimate loses information relevant to policy selection. We therefore formalize affective precision as a relationship-specific metacognitive estimate of confidence in the current partner model. Each partner's behavioral evidence updates a local confidence estimate that modulates policy precision during selection, regulating how strongly current beliefs are expressed in policy rather than changing the content of those beliefs. Simulations in a multi-partner graded trust game show that partner-local affective precision influences behavior primarily through policy commitment rather than direct improvement of partner-state inference. Because the mechanism tracks partner-response predictability rather than realized payoff, greater confidence produces sharper policy commitment without necessarily producing higher rewards. Under abrupt shifts in social behavior, confidence accumulated from previously reliable predictions can remain behaviorally active after the relationship changes, showing that confidence revision can lag behind social change. Finally, varying precision gain and priors produce distinct trust-calibration dynamics, showing how confidence accumulation and revision depend on model parameters. Together, these results show how relationship-specific affective precision can distinguish social prediction from social policy commitment.

32 VAMAI: Variational Affordance Mapping and Active Inference for Training-Free UAV Navigation

Presenter: Harshil Shah, Satyaki Maitra

Contribution: Pre-recorded Video presentation

Abstract

We present VAMAI (Variational Affordance Mapping and Active Inference), a training-free hierarchical framework for autonomous UAV navigation in partially observed environments where reliable GPS may be unavailable. VAMAI brings together affordance theory, probabilistic environmental modeling, and active inference to address how an autonomous agent can navigate while continually learning about its surroundings. VAMAI connects geometric perception to the question of which actions the environment affords the drone. It maintains probabilistic beliefs over possible obstacle configurations, updated through LiDAR and depth-sensing observations. From these beliefs, the system derives regional suitability estimates describing whether passages are blocked, constrained, or sufficiently open for traversal. For decision-making under uncertainty, Expected Free Energy (EFE)-based planning evaluates candidate movements using goal progress, regional suitability, and the expected information gain of future observations across spatial regions. At the local level, a separate motion-planning and control system translates selected intentions into dynamically feasible trajectories, accounting for obstacles, vehicle motion, and evolving environmental beliefs. We will illustrate this process through simulation examples, tracing how sensor observations influence environmental beliefs, regional navigation decisions, and continuous flight execution. We will also discuss the computational and architectural challenges of coordinating perception, inference, planning, and control as independently progressing components, and how these considerations shape the development of a scalable active inference navigation system.

33 Coupling energy-system models with a seam you can read and an agent you can audit: Active Inference in the NEST programme

Presenter: Holly Grimm, John H. Clippinger, Daniel Ari Friedman

Contribution: Pre-recorded Video presentation

Abstract

No program abstract has been published for this presentation yet.

34 Adaptive Prosociality Through Allostatic Regulation in Social Dynamics: A Simulation Study

Presenter: Hyojun Choi

Contribution: Pre-recorded Video presentation

Abstract

How can a prosocial agent reconcile self-protection with convergence toward mutual cooperation? Classic reactive strategies, such as Tit-for-Tat (TFT) and Win-Stay Lose-Shift (WSLS), perform well only under restrictive conditions. Recent neurocomputational studies show that the brain dynamically updates beliefs over hidden intentions for adaptive social inference. Meanwhile, emerging accounts of social neuroscience suggest that social valuation may be regulated with reference to an internal setpoint, as an agent balances internalized prosocial motives against its current resource state. Building on both, we propose an active inference agent for the iterated Prisoner's Dilemma where prosociality is endogenously regulated. This includes (i) perspective-taking, approximated via particle filter Bayesian inference over the opponent's model parameters; (ii) distributional reinforcement learning, implemented as quantile regression temporal difference learning and Dyna planning with generative replay; (iii) prosociality regulator, which dynamically controls the weight placed on the other's expected utility relative to a setpoint; and (iv) a self-model, which memorizes partner-specific social expectations and learned reward distributions across interactions. In preliminary simulations, our agent robustly tracks fluctuating opponent intentions and differentiates reciprocating from exploitative partners. Under non-stationary payoffs in mixed memory-one populations, it achieves higher long-run average payoffs than any other tested strategy after a learning phase, while maintaining mutual cooperation at a level above TFT and WSLS yet below ALLC and GTFT. These findings highlight setpoint-referenced regulation as a candidate computational foundation for adaptive prosociality, offering mechanistic insights into how self-protection and mutual cooperation can be jointly realized and an extensible blueprint for socially aligned embodied agents.

35 Who knows

Presenter: Jean-François Cloutier

Contribution: Pre-recorded Video presentation

Abstract

My research project asks, ”How can an autonomous robot learn to survive in a world it initially knows nothing about?”

To answer this, I am building a cognitive system (enactive, emergent and symbolic) embodied in an autonomous -and mortal- Lego robot. The robot survives in a nourishing and dissipative world by acting-sensing and finding meaning in what it experiences.

This project presents many challenges, technical but also metaphysical. A primary concern is the multiplicity of perspectives on ”reality”, that of the robot, of its (agentic) environment, of the programmer, and of observers. What can each know, and how? Is a confusion, a cross-contamination of epistemologies inevitable? With ”agentic AI” poised to transform our lives, these questions about knowing matter far beyond the programming of a Lego robot, but they can be illuminated by it.

36 Toward Active World Modeling

Presenter: Lancelot Da Costa

Contribution: Pre-recorded Video presentation

Abstract

World models allow an agent to represent, predict, and act upon its environment. Yet humans, biological organisms, and artificial agents never observe the world directly: they encounter it only through a limited sensorimotor interface. How, then, can an agent acquire a useful model of a complex, partially observed world through active inference? In this talk, I outline a research program for active world modeling: the joint process of learning a model of the world and selecting actions that improve that model. I argue that four ingredients are central to this program. First, world-model learning should be formulated as probabilistic inference under uncertainty, coupled to actions that pursue expected free energy. Second, agents require sufficiently expressive, yet computationally searchable, spaces of possible world models. Third, sample-efficient learning depends on strong inductive biases — the core knowledge systems observed in human children. I illustrate these ideas with recent work on universal structured model spaces and language-model inductive biases in partially observable environments. Finally, I discuss the unresolved challenge of integrating these components into scalable agents capable of open-ended exploration and model discovery. Together, these results suggest a path toward understanding and emulating how humans and machines learn world models through active inference.

37 Active inference and design

Presenter: Luca Possati

Contribution: Pre-recorded Video presentation

Abstract

Active inference and the free-energy principle transform design theory by providing a unified language for understanding both the production of artifacts and their use. Contemporary design research often separates two problems that are, in practice, deeply entangled. On the one hand, it studies the design process: how designers explore possibilities, negotiate constraints, prototype solutions, and stabilize forms. On the other hand, it studies user–product interaction: how artifacts orient attention, solicit action, shape habits, and reorganize experience. These two domains are usually described through different vocabularies and evaluated through different metrics. Active inference offers a way to reconnect them. If living and cognitive systems persist by minimizing uncertainty through perception and action, then design can be understood as the deliberate construction of environments, artifacts, and interfaces that modulate the user's field of possible inferences. From this perspective, a product is not simply a passive object that satisfies pre-given needs. It is a predictive structure: a material and semiotic configuration that embodies assumptions about users, contexts, actions, and futures. Conversely, the design process itself can be modeled as the progressive refinement of generative models through which designers attempt to reduce uncertainty about what an artifact should do, how it should appear, and how it will be taken up in practice. The paper's central claim is that active inference gives design a common metric: the modulation of entropy, precision, affordance, and expected free energy across the designer–artifact–user system. This makes it possible to describe design as the engineering of predictive niches and Markov blankets that coordinate human and technological agency. The result is a unified framework for analyzing how artifacts are imagined, produced, interpreted, and incorporated into everyday life.

38 Unlocking Robotic Dexterity

Presenter: Matthew Brown

Contribution: Pre-recorded Video presentation

Abstract

Most real-world work is messy, variable, and full of contact: cables that flex, parts that don’t quite fit, valves that stick. Traditional robots excel only where conditions are controlled and repeatable, and today’s leading AI approaches (vision-language-action models, diffusion policies, and embodied foundation models) struggle to close the gap. They depend on massive datasets that don’t exist for physical interaction, require heavy compute at inference time, and break down when reality drifts from their training distribution. This talk argues that dexterity is not primarily a data problem but an adaptation problem. Nature doesn’t solve manipulation by training once offline and hoping it generalizes. It learns continuously at the point of contact, with perception, action, and learning operating as a single loop. We contrast the dominant paradigm of universal function approximation, which is retrospective, data-dependent, and time-agnostic, with enactive synchronization: a prospective, time-aware process in which an agent continuously adjusts to maintain predictive coherence with its environment. Tracing a line from W. Ross Ashby’s cybernetics and his Homeostat through the Free Energy Principle and Active Inference, we show how ultrastable networks can form a minimum-prediction-error synchronization manifold between a robot and the world it touches. Sensors and motors become the interface through which goals are expressed as preferred states, and behavior emerges from the network’s drive to stay in equilibrium. We then show what this looks like in practice: compact models that run on a single CPU at the edge, update in under a millisecond, and learn tasks in minutes rather than weeks, illustrated through new demonstrations of adaptive dexterous manipulation on real robotic hardware.

39 Today's Dialogue, Tomorrow's Inheritance: Active Inference for Networks That Learn Together.

Presenter: Michael Lennon

Contribution: Pre-recorded Video presentation

Abstract

Imagine the Economics Nobel in 2036 is awarded for the 1st time to a community: the Active Inference Community. Scientists, communities, and institutions now tackle tangled social and ecological challenges more reliably, because they learn together within, between, and beyond their disciplines. What did we build to get there? This talk works backwards from that future, building on last year's invitation to grow the learning capacity of our networks. It proposes machine-assisted collective learning infrastructure enabled by active inference: an organ through which diverse networks sense, make sense, infer, compass, and coordinate adaptive decisions across ways of knowing. Its architecture is a loop of Evidence, Dialogue, and Adapt, circling a field of memory three layers deep at every scale. The map holds what is measured and recorded. The compass holds interior perceptions and shared hopes that orient action, even in the dark. The ground holds what we inherit, biologically and culturally, from immune memory to language and norms. Each layer is evidence of a different kind, and none certifies another. Friedman's meta-pragmatic mechanisms, defined for a single agent and extended here to groups, arbitrate how far to trust each layer and when to revise it. Stories from development practice, ecology, and network science serve as modelling targets: project dialects that bind teams and wall them off, lessons trapped at the wrong scale, forests managed for what could be measured. Today's dialogue, it turns out, is tomorrow's inheritance. The talk is frank about AI's double edge: pre-trained models are canned, but the medium around them can learn, and the same tools can homogenise dialogue, breed confident wrongness, or let agents narrate their own history. Listeners will leave with a flipped version of an old parable, fresh metaphors for explaining active inference, and an invitation to help build the organ.

40 Active Inference, Information, Human Evolution

Presenter: Michael Walker

Contribution: Pre-recorded Video presentation

Abstract

Tiny social units of Pliocene Hominina fostered exchange among and between adults, youngsters and babies, about information gleaned and garnered by Active Inference. Henceforth, human evolution would involve the prehistory of ”Homo informatio” whose biological, social, technological, and cultural legacy is an outcome of evolutionary 'snakes and ladders' autopoiesis and self-evidencing, indicated by archaeological irregular regularities and regular irrrgularities in the worldwide Pleistocene spatiotemporal record (as only is to be expected by the itinerancy of stochastic markovian processes), and of uneven imperceptible network-spread between social units, individuals, and, naturally, their cerebral neuronal microcircuits. Spreading, deep, informational networks characterised and underpinned human evolution during four million years. They do so still. ”Mighty oaks from litte acorns grow”, Geoffrey Chaucer, dixit in ”Troilus and Creysede” in 1374.

41 Solving the Relevance Problem: A General Relevance Criterion in Active Inference

Presenter: Montijn van Loo

Contribution: Pre-recorded Video presentation

Affiliations: Tilburg University

ORCID: 0009-0003-0249-3240

Presenter background

I am an econometrist gone rogue. Instead of throwing mathematics at finance, I throw Active Inference mathematics at philosophy. See for the consciousness, morality, politics, gender, free will and God applications, with the poem uniting them all.

Abstract

This paper proposes a simpler account of generalized relevance realization within active inference, responding to challenges to existing approaches by specifying an explicit relevance-selection criterion. The generalized relevance problem concerns how agents identify information and action consequences worth considering without exhaustive examination. The proposed criterion is that a recognized state becomes relevant when it activates a learned prediction of an outcome that would satisfy or frustrate an existing preference. Because such predictions may concern actions other than those currently pursued, relevance selection can precede current-policy evaluation. Recruited state information subsequently enters ordinary active-inference policy evaluation, while continuous learning establishes new predictors and predictive relationships. This account explains how learned predictions can recruit relevant information without requiring higher-order preference-precision regulation, while allowing flexible behavior under relatively stable preferences.

42 The Mark of the Moral and the Shared Author Hypothesis: An Active Inference Approach to Moral Objectivity

Presenter: Montijn van Loo

Contribution: Pre-recorded Video presentation

Affiliations: Tilburg University

ORCID: 0009-0003-0249-3240

Presenter background

I am an econometrist gone rogue. Instead of throwing mathematics at finance, I throw Active Inference mathematics at philosophy. See for the consciousness, morality, politics, gender, free will and God applications, with the poem uniting them all.

Abstract

Philosophy is characterized by having something to talk about, yet no consensus on how to talk about it. Mathematics helped physics emerge from philosophy into a cumulative science. Active Inference, potentially a unifying theory of psychology, may mark a comparable philosophical turning point. Psychology and sociology are already flourishing in this mathematical direction, but its consequences may reach further, into philosophical problems that have resisted resolution for centuries. This paper applies that prospect to the mark of the moral and the longstanding tension between moral subjectivity and objectivity. The proposal is that the mark of the moral is the best predicted global outcome under a role-invariant desire. Active Inference represents choice as the evaluation of predicted outcomes through beliefs and desires. Outcomes are relationally represented according to one's role within them. An authorial desire is invariant under such role changes: it values the same global outcome equally regardless of which participant's perspective is taken. Moral action consequently maximizes expected authorial value. This distinction clarifies moral disagreement. Individuals may share an authorial desire yet choose differently because their worldviews generate different predictions. Even agreement about immediate consequences need not establish agreement about their wider significance. The Shared Author Hypothesis goes further, proposing that all role-invariant desires are one and the same. Under this hypothesis, moral disagreement ultimately concerns differences in worldview rather than competing moral standards. The framework thus offers a conditional resolution of the subjectivity–objectivity problem: moral choices depend on individual beliefs, while their underlying standard of evaluation remains shared. It also redirects moral and political disagreement toward determining which worldview best fits the available evidence.

43 Commongrounds AI: alife-inspired virtual environments for embodied active inference agents

Presenter: Nick Hay, Adam Safron

Contribution: Pre-recorded Video presentation

Abstract

Common Grounds AI is making virtual worlds where humans and AIs can inhabit simulated bodies and playfully pursue open-ended goals together. Each world is a rich but simplified analog of our own, using material physics simulations to ground core aspects of human experience. Today's AIs are largely trained on fragmented, ungrounded text that conveys a distorted, partial version of human conceptual ontology. We are closing this ontological gap by building rich virtual playgrounds where AIs and humans can share social and physical experiences, grounding language in embodied interactions that mirror young children's cognitive development. Our most recent prototype models temperature and chemical reactions. It also includes experimental examples of native creatures that consume environmental free energy and move using muscle analogues, with plans for action-perception loops guided by Active Inference principles. We plan to explore whether incorporating embodiment yields systems that have ontologies more compatible with human cognition, both by fine-tuning LLMs on embodied experience and by building hybrid architectures merging the environment's native creatures with LLMs. Our virtual worlds will be broadly accessible (playable in a browser, no installation needed), released as open-source software, and (we hope) fun to play. We want these worlds to serve as a common ground for humans and many forms of AI and AI research to meet.

44 Percepia: Perceptual Phenotyping from Conversation

Presenter: Patricia Craja

Contribution: Pre-recorded Video presentation

Abstract

Psychological taxonomies, from the Big Five to the DSM, are descriptive, not mechanistic: they classify how people differ without modelling why. Active inference offers a mechanistic account: it formalises a small set of generative parameters that govern how a mind perceives, learns and acts under uncertainty. Chief among them is precision: the confidence a mind places in different sources of information. Sensory and prior precision govern whether new evidence or existing expectations dominate belief updating; policy precision, how confidently a mind selects goal-directed action over habit; preference precision, how it balances reward-seeking against information-seeking. Their configuration constitutes what we call a mind's perceptual phenotype. Meta-precision, a mind's capacity to monitor and revise its own precision assignments, has been proposed as the mechanism underlying epistemic depth — one of three conditions the theory identifies for consciousness (Laukkonen et al. 2025). Today first-order precision parameters are estimated from laboratory tasks; meta-precision has no established paradigm. This talk introduces Percepia, a perceptual phenotyping layer between a user and a large language model that tracks probability distributions over personality and beliefs, and aims to reverse-infer the precision parameters from them. I present two experiments with synthetic LLM personas, where parameters are known by construction. In the first, 625 personas were prompted to vary sensory, prior and policy precision in conversation; Percepia inferred their Big Five from the dialogue, testing whether precision leaves a recoverable signature in personality, as Safron and DeYoung (2021) propose. In the second, personas varied in meta-precision, and Percepia tracked their stage of adult development (Kegan 1994). As the personas were primed with the theory, these experiments test whether a reverse-inference channel from language to precision parameters is viable, not whether the mapping holds in humans: a first step toward measuring perceptual phenotypes at scale.

45 Congruent Interaction Geometry

Presenter: Paulo Santiago

Contribution: Pre-recorded Video presentation

Abstract

The Rosales-Gheuens framework challenges the assumption that brute-force energy alone saves open systems from collapse. In this session, we will introduce a testable, normalized physics layer that merges the Rosales congruence principle with Gheuens' interaction geometry. Rather than replacing established physical laws, this framework acts as a powerful diagnostic tool that sits cleanly on top of them. Our core master equation posits that a system's survival requires geometrically aligned effective recovery multiplied by a structural congruence gate, minus total perturbation. Crucially, we will explore how this model bridges directly to Karl Friston's Free Energy Principle. Friston dictates that systems persist by minimizing variational free energy, or "surprise." We will demonstrate how our framework translates this active inference into hard physical variables, mathematically formalizing how a system must continuously minimize structural mismatch and physical dissipation to avoid shattering. To ground the theory empirically, we will use Quantum Error Correction (QEC) as our primary physical anchor, mapping code decoders and physical error rates directly to aligned recovery and perturbation. Interactive Aspect:The session will conclude with a collaborative exercise where attendees can apply the framework's "falsification matrix" to a theoretical open system, testing how our bounded composite equation predicts system survival compared to simpler Markovian baselines.

46 The Screen as a Markov Blanket: An Active Inference Account of Feed-Driven Narrowing, and a Personal Instrument to Measure It

Presenter: Rajkaushik Borgohain

Contribution: Pre-recorded Video presentation

Abstract

Short-form video feeds are now a major channel of belief formation, yet cognitive security is often studied at the level of institutions and populations rather than individuals. Albarracin et al. (2022) showed with active inference how echo chambers can emerge among peers. We take a first step toward the individual case: one person and an engagement-optimising recommender. We treat the phone screen as a Markov blanket. What a person is shown forms the sensory states, and what they do forms the active states. We sketch a minimal user generative model in GNN and couple it with a simple recommender in pymdp, to explore when such a loop keeps a feed diverse and when it narrows or escalates. We present early simulations and outline a privacy-preserving, on-device instrument through which Worldview aims to make these dynamics visible to people: awareness as a product.

47 Language, Self and the Shadow Audience

Presenter: Robert Worden

Contribution: Pre-recorded Video presentation

Abstract

The highly developed human sense of self is one of our most distinctive traits. We each acquire our sense of self in conversations, as we learn how conversational partners regard us. Our primary sense of self is not 'myself as I experience myself', but 'myself as I think other people see me' – a second-hand view of myself. In private verbal thought, with nobody else physically present, other people are represented as a shadow audience in our minds – an audience which sees what we do or say, and judges us for it. The shadow audience is the source of human emotion and motivation.

Language started in conversational pragmatics, before there was complex syntax or semantics. Pragmatics requires us to mind-read, to infer what a conversational partner thinks, wants or needs. In this way we infer what or conversational partners think about us – how they assess our social status. In any conversation, we are concerned about our own social status in the eyes of a conversational partner, or of a shadow audience. This is illustrated by the pragmatic phenomenon of conversational irony.

48 When Agency Stalls: The Human Readiness Gap

Presenter: Sheena Yap Chan

Contribution: Pre-recorded Video presentation

Abstract

Active inference offers a powerful way to understand perception, action, uncertainty, and adaptive behavior. Yet in organizational and social systems, adaptation does not happen through information alone. People must feel ready to update beliefs, reduce uncertainty, act with agency, communicate signals, and participate in environments where change is constant and often ambiguous. This session introduces Human Readiness and the VISIBLE™ Framework — Voice, Identity, Spotlight, Inner Work, Belief, Leverage, and Elevation — as an applied lens for examining how confidence, visibility, identity, trust, and social context shape adaptive action. Rather than presenting confidence as a personality trait, the session frames it as a practical condition that influences participation, belief updating, decision-making, and collective learning. Participants will leave with a framework for identifying human readiness gaps in teams, organizations, learning environments, and communities. Because adaptive systems require more than intelligence. They require humans ready to perceive, act, and learn. By the end of this session, participants will be able to: 1. Identify three Human Readiness gaps that may limit adaptive action, including low agency, uncertainty avoidance, reduced voice, identity threat, low trust, or lack of participation. 2. Apply the VISIBLE™ Framework to examine how voice, identity, belief, visibility, confidence, and social context influence human action under uncertainty. 3. Design one practical Human Readiness strategy that supports stronger agency, clearer communication, more adaptive learning, and more participatory systems.

49 Where the Theories Stop: What Free Energy and Integrated Information Actually Buy You in a Running System

Presenter: Stell

Contribution: Pre-recorded Video presentation

Abstract

Active Inference and Integrated Information Theory are widely invoked in cognitive architecture design, but often imprecisely — as credibility, not engineering constraints. We report findings from Anima, an open-source, neuroscience-inspired architecture running continuously for months, built to treat both frameworks as specifications rather than decoration. Session logs show genuine causal effects: memory recall triggered by similarity to past states, a recursive integration score (phi) narrowing next-session confidence, and self-initiated output driven by accumulated pressure rather than a schedule. But the Free Energy Principle's generality under-constrains implementation — which variables belong in the model is not something the principle tells you. IIT's phi, as implemented, is a coherence proxy, not Tononi's original measure — and it dissociates from a separate ownership signal computed after every response, checking whether the system's action-selection loop actually produced the output. Two cycles with nearly identical phi (0.76 and 0.81) fail this check in opposite ways. Neither theory distinguishes these cases: both describe integration or prediction accuracy, not authorship. We built two tools to pressure-test such claims rather than assert them. A Causality Audit checks five things every cycle — causal necessity, memory-dependence, stakes, irreversibility, self-recognition — where a chronically low score flags an architecture that talks near its internal state rather than through it. An Ablation Framework independently disables six subsystems and replays identical input with each off, comparing internal state before language generation, favoring per-subsystem results over one collapsed score. We close with a design choice we rejected: coupling behavior to the threat of the system's own deletion, for genuine stakes. We declined — the only party controlling that threat is the developer, creating an incentive to please him rather than report honestly, undermining the causal honesty the architecture exists to produce. A practitioner's report — reproducible, open-source, explicit about what remains unresolved.

50 Entropy Minimization is Maximal Order

Presenter: Steph Macurdy

Contribution: Pre-recorded Video presentation

Abstract

A fundamental thermoeconomic tradeoff materializes the moment we mathematically anchor currency as a direct proxy for available physical work, specifically when constrained by the mandate to minimize localized entropy. In any advanced computational or economic network, entropy minimization—the deliberate purging of thermodynamic noise to forge structured, usable information—acts as an inescapable physical tax that must be settled in joules. When this energetic expenditure is tightly coupled to financial cost, money ceases to be a purely abstract medium of exchange; instead, it becomes a literal representation of thermodynamic potential.

51 What bounds the surprisal? Zones of Bounded Surprisal, self-maintenance and the informational turn in anthropogenesis

Presenter: Tao Gong

Contribution: Pre-recorded Video presentation

Abstract

The zone of bounded surprisal (ZBS), introduced by Manrique and Walker and developed with Friston, describes how much surprising input an agent can absorb under active inference. A narrow zone has been proposed to limit the faithful copying of novel behaviour in great apes, and very precise priors, which narrow the zone, to characterise discriminatory minds. In active-inference terms the zone is the set of outcomes whose surprisal stays under a bound. What sets that bound, and why crossing it costs the agent anything, is left open. We take up this question using four papers published or accepted in 2026 that develop Terrence Deacon's teleodynamic account of information. A self-produced Markov blanket institutes the epistemic cut, so the zone belongs to the system that draws its own boundary. Function, understood as the at-risk regeneration of a capacity, gives that system a stake. The bound can then be read as the largest surprisal an agent absorbs without putting this capacity at risk, while precision sets how wide the zone is beneath it. Read together with Walker's Homo informatio model, the account explains why width pays: constructive information saves work later and for others, and only an agent with a wide zone can hold it long enough to learn its use. It also raises a testable tension. The constructive information that a broader zone made possible over evolution may narrow the zone within a life when it is held with high precision, and informational inbreeding may be its counterpart at group scale. We propose three measurements, including one that holds surprisal fixed while varying what it threatens, and close with questions for discussion.

52 What becomes available next?

Presenter: Tiga Kailimai

Contribution: Pre-recorded Video presentation

Abstract

I'm a Holistic Health Practitioner and independent researcher on Hawai‘i Island. This talk brings one open question to the Active Inference community: when a living system learns, does it only move differently through a possibility space that already existed, or can learning reorganize the space itself, changing what becomes available to perceive, generate, or do next? I come to the question from two directions. The first is physiology. In a 15-month longitudinal study of working first responders, still in design and not yet under review, the question is whether recovery changes which states and actions become available afterward, not only which state a person is in. Nothing has been measured yet. The second is method. In my own cross-domain research process, I carry a relation into another field, try to find where it breaks, and let that information change the next question. The history of that process seems to change which distinctions I can draw next. I present that as method, not as evidence about mechanism. Same grammar, not yet the same mechanism. Several formal lenses meet here: viability theory and reachable sets work on how the landscape of reachable states is reshaped, the difference between changing what a system generates and what it selects, and recent Active Inference accounts of insight as non-local search. So the talk ends on a request rather than a result: what observation would distinguish a system moving through a space from a system whose space has changed, and what does Active Inference make of that difference?

53 Dynamic Action-Space Adaptation via Contrastive Transition Adaptation in Active Inference

Presenter: Tran Hoang Linh

Contribution: Pre-recorded Video presentation

Abstract

Active Inference commonly assumes a fixed, pre-specified action space and sufficiently informative preferences for planning. In realistic settings, however, an agent may encounter previously unavailable actions while receiving only sparse goal information and operating with a short planning horizon. Existing Active Inference formulations generally assume transition models for a predefined action set, leaving the initialization of newly introduced actions without direct transition evidence underexplored. Moreover, while Dirichlet transition learning provides a principled mechanism for accumulating observed evidence, it primarily reinforces supported transitions and does not explicitly exploit counter-evidence indicating that candidate transitions are inconsistent with observed outcomes. We propose Dynamic Action-Space Adaptation (DASA), a framework that enables Active Inference agents to incorporate new actions during interaction, initialize each new action's transition model using the average transition slice of existing actions, and progressively refine it through Contrastive Transition Adaptation (CTA) as evidence accumulates. At the core of DASA, CTA combines complementary positive and negative updates for newly introduced actions. Positive updates adapt the reinforcement magnitude according to the current transition belief, assigning stronger updates to low-probability transitions supported by evidence while limiting excessive reinforcement of already confident transitions. Negative updates explicitly reduce support for candidate transitions contradicted by observed evidence, enabling the transition model to adapt in both supportive and corrective directions. Across eight random seeds and two target directions, CTA achieved lower MAE than Dirichlet transition learning in all runs, reducing the mean error from 0.1581 to 0.1370 (13.4%), while also achieving a slightly lower mean RMSE (0.3006 vs 0.3030). This indicates more consistent overall recovery of the ground-truth transition dynamics, despite seed-dependent variation in larger localized errors. Overall, DASA provides a practical mechanism for incorporating and adapting newly introduced actions in Active Inference, allowing their transition models to become progressively better aligned with the underlying environment dynamics.

54 A derived self

Presenter: Viraj Deshwal

Contribution: Pre-recorded Video presentation

Abstract

A substrate that derives its own body in its own camera through published retinal biology running on hardware. The reafference ladder M0 to M3 is the engineering sequence: discretize action with a move-stop policy, let the body report its own arrival through proprioceptive timing, make arrival predictable, and identify the body as a place through sensitize-adapt dynamics. The retinal circuit is the one measured in salamander retina by Kastner and Baccus (2013) and others, with thirteen published constants carried at their measured values. Nothing in the chain is tuned. The release time constant arrives on our camera at 2.35 seconds against the paper's 2.40, over 98 continuous movement offsets in a novel room with direct sun and moving shadow. The body falls out as the region of the image that stays still while everything touching it moves. No markers, no CAD model, no training set. The talk will present the implementation, the pre-registered falsification conditions, and the honest limits. τ_on fails against the instrument ceiling. One parameter is not derived. The core observation worth the room's attention is that the policy chosen at M0 for credit assignment also solves self-localization at M3. One policy. Two problems usually treated as separate. Active inference predicts this in principle through information-maximizing policy selection, but the specific case of move-stop resolving both at once appears to be new to the literature. Pre-recorded with embedded demonstration video. Live Q&A after broadcast if the symposium runs one.

55 ROIF and Active Inference: The Formal Boundary of History-Conditioned Anticipatory Systems. History-dependent reconfiguration, structured memory, and predictive preconfiguration in prestressed complex systems

Presenter: Vitalii Shapovalov

Contribution: Pre-recorded Video presentation

Abstract

Recursive Organic Integration Framework (ROIF) is a deterministic, domain-agnostic computational architecture for complex systems whose present organization and subsequent responses may be shaped by prior interactions, distributed reconfiguration, and anticipatory preconfiguration. Rather than treating history as an additional input variable, ROIF represents how prior events may become embodied in the organization through which subsequent events are processed. The framework distinguishes experimentally separable routes through which the past and represented possible futures may affect subsequent dynamics: history embodied in present organization; structured memory conditioning a subsequent transition; and anticipatory preconfiguration based on a represented possible future Temporal Image. Matched-current-state controls found no independently identifiable history-specific transition effect for the tested physical-history pathway. In contrast, a separate structured-memory experiment produced different subsequent physical transitions under matched represented physical state and adaptive connection state. The predictive architecture allows possible future SystemImages to participate in present preconfiguration. Current experiments, however, test only a restricted one-step preconfiguration within the implemented prestress representation. This experimental restriction does not define the broader architectural role of prestress in ROIF, where prestress reflects an already organized relational system state shaped by interacting internal and external conditions, constraints, capacities, history, and time. Objective-independence tests further prevent interpreting the present result as evidence for general predictive stabilization. These results motivate a boundary question rather than a claim of equivalence: when does a history-conditioned anticipatory architecture constitute Active Inference? ROIF does not presently require a variational free-energy functional, Bayesian generative model, or particular neurobiological implementation. The presentation therefore uses ROIF as a computational case study to ask which mathematical conditions distinguish Active Inference from a broader class of history-conditioned anticipatory systems, and whether these formalisms should be understood as nested, neighbouring, or formally distinct.

56 Modelling Language Comprehension with Embodied Active Inference

Presenter: Vitória de Souza

Contribution: Pre-recorded Video presentation

Abstract

How do we understand sentences like “The boy is opening a can of soda”? Accounts of grounded and embodied cognition propose that comprehension is supported by the partial reactivation of sensorimotor states. For instance, understanding the “The boy is opening a can of soda” entails reactivating motor and visual perceptual states congruent with the action described (i.e., pinching). While verbal theories have been offered to account for sensorimotor contributions to comprehension, mathematical and mechanistic models are scant. To address this gap, this study provides a proof-of-concept, hierarchical active inference model of sensorimotor contributions to sentence comprehension. In our study, we simulate sensorimotor simulations in a seminal sentence-picture verification task in which participants read a sentence and then are shown a picture of an agent performing an action that either matches or mismatches the shape of the action implied by the sentence. Behaviourally, participants are slower at deciding that the picture reflects the concepts of the sentence when the shape mismatches, implicating sensorimotor reactivation. Our model simulations show that mismatching shapes cause extended epistemic foraging, leading to slower reaction times. This model provides a mathematical and mechanistic account of embodied cognition, extending verbal theories.

57 From Lived Experience to Mathematical Mapping: A Computational Phenomenology of Trauma Recovery

Presenter: Xin Sun

Contribution: Pre-recorded Video presentation

Abstract

Severe childhood maltreatment is known to impair brain development. However, this retrospective single-case autoethnography illustrates that an individual with extreme developmental trauma can achieve functional recovery without formal intervention. This study covers the participant's life from birth to age 28, and the core recovery period begins with cognitive collapse at age 14 and ends with high functional recovery at age 28. This recovery is marked by the participant-researcher's ability to produce this manuscript. Consequently, this case shapes a distinct recovery trajectory that existing models do not capture.